

Earth and Environmental Sciences: Open Data, Open Source Code

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Outline

1 Open Research

2 Open Data

3 Open Source Code

4 Data Assimilation / Machine Learning

Research Topics

- ▶ enter your ideas
- ▶ and so on

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Satellite Data

Satellite data are a public good

- ▶ For everyone
- ▶

Anything bad?

- ▶ Maintenance, misinterpretation
- ▶

Outline

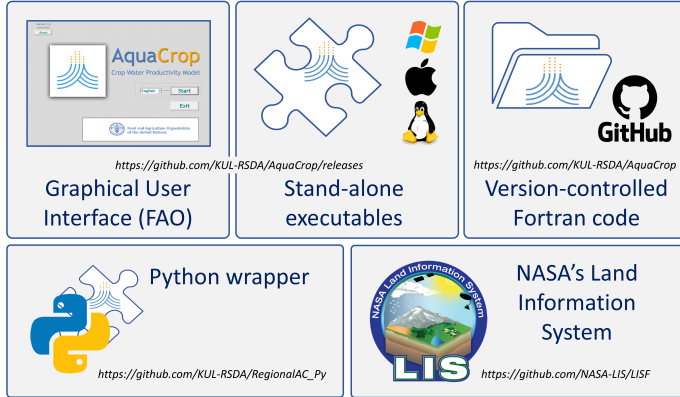
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AquaCrop Model



(De Lannoy et al., 2026)

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Data Assimilation

"The whole is greater than the sum of the parts" (Aristotle)

(De Lannoy et al., 2022)

Conclusions

Serve the world

- ▶ science is a global public good
- ▶ open satellite data
- ▶ open source code
- ▶ open access
- ▶ give and take

Some references I

- De Lannoy, G., Busschaert, L., Bechtold, M., Lanfranco, N., de Roos, S., Heyvaert, Z., Mortelmans, J., Scherrer, S., van den Bossche, M., Kumar, S., Mocko, D., Fereres, E., Garcia-Vila, M., Heng, L., Steduto, P., & Raes, D. (2026). Advancing crop modeling and data assimilation using aquacrop v7.2 in nasa's land information system framework v7.5. *Geoscientific Model Development*, 19(6), 2551–2575.
URL <https://gmd.copernicus.org/articles/19/2551/2026/>
- De Lannoy, G. J. M., Bechtold, M., Albergel, C., Brocca, L., Calvet, J.-C., Carrassi, A., Crow, W., de Rosnay, P., Durand, M., Forman, B., Geppert, G., Girotto, M., H.-J. Hendicks Franssen, Jonas, T., Kumar, S., Lievens, H., Lu, Y., Massari, C., Pauwels, V., Reichle, R., & Steele-Dunne, S. (2022). Perspective on satellite-based land data assimilation to estimate water cycle components in an era of advanced data availability and model sophistication. *Frontiers of Water*, 4.
URL <https://www.frontiersin.org/journals/water/articles/10.3389/frwa.2022.981745>